

YUHAN MU

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EDUCATION BACKGROUND

Beijing University of Chemical Technology

Rank 7/80, GPA 3.71/4.33

Mechanical Engineering, *undergraduate*, 2024.9-Now

Scholarship: People's Scholarship, Extracurricular Achievement Scholarship

SKILLS

- Python, C++, ROS 2 Humble, PyTorch, YOLOv8, OpenCV, Gazebo, RViz, Inverse Kinematics, STM32, CVAT, Git, Linux

EXPERIENCES

Robomaster University League | Competition, Vision Group Nov. 2025-Mar. 2026

- Overview: Built a real-time vision-guided auto-aiming system for sentinel robot, integrating Open CV, Yolo detection, and embedded control into a perception-to-actuation pipeline deployed in competition.
- Method: Migrated rm-vision method to HikRobot industrial camera under ROS 2. Completed serial communication protocol for 1kHz vision-to-gimbal commands on STM32 firmware. Calibrated dataset and trained YOLOv8n detector with an $mAP@0.5%$ of 98.7%; Also leveraged real-time OpenCV armor-detection using contour-based morphological filtering to achieve higher-precision candidate. Implemented PnP pose estimation adjustment with multi-hypothesis tracking.
- Auto-aiming test video: <https://www.bilibili.com/video/BV111ut6mEnq/>

RoboCup@Home | Competition, Team Member Mar. 2026- May. 2026

- Overview: Enhanced an autonomous navigation system for a service robot by fusing vision-based detection with Nav2 path planning.
- Method: Annotated household object dataset with CVAT and trained YOLOv8n detector. Developed ROS 2 system integrating Nav2 BasicNavigator for multi-waypoint patrol, implementing state machine for sequential navigation-detection cycles to provide real-time feedback which contained vision results, navigation and voice announcements.
- GitHub repo: <https://github.com/Christinemiu/2026-Robocup-Vision->

RAICOM Robot Developer Competition | Competition, Team Leader May. 2025-Jul. 2025

- Overview: Achieving periodic squatting and standing-up motions of a quadruped robot dog through real-time joint control and multi-leg coordination algorithms based on ROS 1.
- Method: Implemented 19-DOF joint-space trajectory control with linear interpolation at 100Hz, enabling smooth squatting motions under kinematic constraints. Developed complete ROS packages with custom message/service interfaces and real-time joint state control.
- GitHub repo: <https://github.com/Christinemiu/ROS1-Robot-Kinematics-Project-Learning>

COMPETITIONS

- The 1st Prize of Provincial Level in TI Cup National Undergraduate Electronic Design Contest Aug. 2026
- The 3rd Prize in RoboCup@Home May. 2026
- The 2nd Prize of Regional Level in Robomaster University League Mar. 2026
- Honorable Mention in the Mathematical Contest in Modeling(MCM) May. 2026
- The 3rd National Prize in RAICOM Robot Developer Competition Jul. 2025

COURSE WORK

- NTU Summer School on Underwater Image Processing (Best Team Award) Jul. 2026
- Robotics Workshop on Arduino based Line-following Car May. 2025